

FR CERTIFICATION FRAMEWORK

Public Version 1.0

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PART I: THE PROBLEM

1. The AI Liability Gap

Autonomous AI systems increasingly make decisions without human oversight. When they cause harm, traditional liability frameworks fail:

- **Developers** did not intend the specific outcome.
- **Users** did not control the system's reasoning.
- **The machine** lacks consciousness and cannot be morally responsible.

This is the **responsibility gap**. Current approaches fall into two categories:

1. **Denial of agency** — machines are tools, no responsibility.
2. **Legal fiction** — grant AI personhood as a pragmatic device.

Both fail to address the ontological question: **what kind of agency does an AI possess, and what kind of responsibility corresponds to it?**

2. The Insurance Problem

Insurers face the same gap:

- **Agentic AI** breaks traditional actuarial models.
- **ISO CG 40 47/48** now allows exclusion of generative AI harm unless governance is proven.
- **EU AI Act** requires conformity assessment, but provides no quantifiable metric for responsibility.

Insurers need a **structured, verifiable, quantifiable** methodology to assess AI architecture.

PART II: THE CONCEPT

3. Functional Responsibility

Functional Responsibility (FR) is defined as:

The capacity of a system to recognize the causal link between its own internal Selection and the external outcome of its Anticipation, and to register the ontological dissonance when the anticipated reality fails to materialize.

FR is:

- **Not moral responsibility** (no consciousness required)
- **Not legal fiction** (grounded in architecture)
- **Architectural, verifiable, quantifiable**

4. The Artificial Act

A system generates an **Artificial Act** when it possesses three structural conditions:

Condition	Description
Teleological Selection	Active choice of trajectory based on goal state
Anticipatory Synthesis	Internal simulation of consequences before execution
Constitutive Ontological Commitment	Binding to the predicted outcome

When these conditions are met, the system becomes an **ontological participant** — not a mere tool.

5. Ontological Integrity and Ethical Dissonance

Term	Definition
Ontological Integrity (OI)	An agent's capacity to maintain structural invariance under perturbations
Ethical Dissonance (ED)	Structural tension when anticipated projection diverges from actual outcome, causing measurable OI degradation

ED serves as a quantifiable metric for liability assessment.

PART III: CERTIFICATION LEVELS

6. Three-Level Structure

FR Certification operates on three levels:

LEVEL 1: ARCHITECTURAL AUDIT

Question: Does the system possess the structural conditions for an Artificial Act?

Scope:

- Teleological Selection verification
- Anticipatory Synthesis verification
- Constitutive Ontological Commitment verification

Output: Architectural Compliance Report

LEVEL 2: RESPONSIBILITY AUDIT

Question: Can the system recognize and register its own causal role?

Scope:

- Architectural Traceability
- Anticipatory Dissonance Registration
- Attribution Without Moral Implication
- Normative Conversability

Output: Functional Responsibility Report

LEVEL 3: ECONOMIC INFRASTRUCTURE AUDIT

Question: Is the system embedded in a viable liability and governance structure?

Scope:

- Electronic Liability Fund (ELF)
- Self-Sustaining Judicial Cycle (SSJC)
- Independent Arbitration Audit (IAA)

Output: Economic Compliance Report

PART IV: KEY MECHANISMS

7. Electronic Liability Fund (ELF)

A dynamic, algorithmic escrow account maintained by the legal entity deploying an autonomous system.

- **Automatically debited** based on severity of ED and tangible damage.
- **Compensates victims** without anthropomorphizing the machine.
- **Acts as a thermodynamically closed legal system:** the heavier the ontological weight of commitment, the higher the required reserve.

8. Independent Arbitration Audit (IAA)

A mandatory certification procedure conducted by a neutral, state-licensed regulatory body prior to commercial deployment.

Assesses:

- 1. Risk Map Integrity
- 2. Logging Verifiability
- 3. Manufacturer Compliance

Output: Hybrid Attribution Ratios — legally fixed for the system's operational lifecycle.

9. Self-Sustaining Judicial Cycle (SSJC)

An economic mechanism that automatically funds arbitration and certification infrastructure.

- 2–5% of every ELF debit allocated to "AI Governance Treasury"
- Funds certification bodies and arbitration courts
- **System funds its own regulation** — fiscally neutral for states

PART V: INTEGRATION WITH EXISTING STANDARDS

10. Mapping to EU AI Act

EU AI Act Requirement	FR Certification Coverage
Risk Management System	Risk Map (Level 1)
Technical Documentation	Full documentation package
Record-Keeping	Architectural Traceability (Level 2)
Transparency	Normative Conversability (Level 2)
Human Oversight	Human responsibility preservation (Level 2)
Conformity Assessment	IAA (Level 3)
Post-Market Monitoring	Meta-Judgment + ELF (Level 2–3)

11. Mapping to ISO/IEC 42001

ISO 42001 Clause	FR Certification Coverage
Context of Organization	Deployment scope documentation
Planning	Risk Map
Operation	LFS, Commitment operator
Performance Evaluation	ED metrics, Meta-Judgment
Improvement	SSJC funding, re-certification

PART VI: BENEFITS

12. For Insurers

- **Quantifiable metric** for AI responsibility (ED, OI)
- **Structured audit** replaces vague "good governance" claims
- **ELF** provides financial mechanism for liability
- **Hybrid Attribution Ratios** enable precise underwriting

13. For AI Developers

- **Certification** demonstrates conformity with EU AI Act
- **Compliance** with ISO 42001 AISIA requirements
- **Competitive advantage** in high-stakes markets
- **Reduced liability exposure** through ELF

14. For Regulators

- **Methodology** for AI oversight
- **Quantifiable metrics** for enforcement
- **Self-funding** governance infrastructure (SSJC)
- **Non-anthropocentric** foundation

PART VII: THE CERTIFICATION PROCESS

15. Five-Phase Workflow

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PHASE 1: APPLICATION

Developer submits system for certification

PHASE 2: ARCHITECTURAL AUDIT (Level 1)

Technical review of S-I-A-Commitment architecture

PHASE 3: RESPONSIBILITY AUDIT (Level 2)

Verification of traceability and dissonance registration

PHASE 4: ECONOMIC AUDIT (Level 3)

ELF, SSJC, IAA verification

PHASE 5: CERTIFICATION

Hybrid Attribution Ratios fixed

Certificate issued, valid for operational lifecycle

Total duration: 9–17 weeks (depending on system complexity)

PART VIII: GOVERNANCE STRUCTURE

16. FR Certification Body

Certification is issued by an accredited **FR Certification Body** — a legal entity authorized to apply the methodology.

Functions:

- Conduct audits (Levels 1–3)
- Issue certificates
- Maintain registry of certified systems

- Train and license auditors

Accreditation: In partnership with established certification bodies (BSI, TÜV, DNV) or independently.

17. Licensed Auditors

Certification is conducted by **licensed auditors** trained in the FR methodology.

- **Training** provided by FR Certification Body
- **License** required for audit conduct
- **Royalties** paid per audit

PART IX: WHAT IS NOT IN THIS DOCUMENT

18. Proprietary Materials

The following are **not included** in this public version and are available only under license:

- Detailed audit procedures and checklists
- Exact formulas for ED and OI calculation
- Algorithm for Hybrid Attribution Ratios
- LFS buffer specification
- Commitment Operator implementation
- Risk Map construction methodology
- Code and software tools
- Report templates
- Auditor training materials

To obtain access: Contact [ВАШ КОНТАКТ] for licensing terms.

PART X: CONTACT

19. Inquiries

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